
PO11Q - Quantitative Political Analysis: From Measurement to Inference

Your Questions, Weeks 1-5

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How are we supposed to differentiate between what is an attribute or a predictor of a concept? it feels very arbitrary

- Yes, because it does depend on the circumstances of the research project, recall the question of economic development vis-a-vis democracy.

Why is it okay to use biased but reliable data, but not the other way around?

- Reliable data consistently produces similar results under consistent conditions, making it dependable for analysis.
- Biased data, while consistently skewed, can still provide useful insights if the bias is understood and accounted for in the analysis.
- In contrast, unbiased but unreliable data lacks consistency, and this inconsistency cannot be corrected.

What is the value in using standard deviation as a measure of spread of data around the mean?

- Descriptive statistics provide a summary of the data
- Standard deviation is a commonly used measure of spread that quantifies the average distance of data points from the mean
- It is useful for understanding the variability of the data and for comparing the spread of different data sets

Why is $n - 1$ used for the calculation of standard deviation?

- We only use $n - 1$ in the context of a sample standard deviation, not for a population standard deviation
- This is because we are estimating the population standard deviation from a sample, and using $n - 1$ (Bessel's correction) helps to correct for the bias that arises from using a sample to estimate a population parameter
- Using $n - 1$ instead of n provides a more conservative estimate of the population standard deviation, as it makes the sample standard deviation larger

What does covariance show?

- Covariance is a measure which assesses the degree to which the values of two variables (X and y) vary together.
- It is defined as:

$$\text{Cov}(\mathbf{x}, \mathbf{y}) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

- More on PO12Q.

Why are we interested in Normal distributions, do these crop up frequently in social science or is it just a matter of simplicity? What even is an abnormal distribution and why aren't we talking about them?

- Normal distributions are important in social science because many natural phenomena and human behaviors tend to follow a normal distribution pattern, making it a useful model for analysis.
- Normal distributions are mathematically convenient and have well-defined properties that facilitate statistical inference, hypothesis testing, and the application of various statistical techniques.
- Abnormal distributions, also known as non-normal distributions, deviate from the characteristics of a normal distribution, such as skewness, and may require different analytical approaches.

What is the difference between z-Values and z-Scores?

- A z-score (or z-value) is the number of standard deviations an observation lies from the mean.
- Both terms are often used interchangeably in statistics to refer to the same concept.
- The z-score is calculated using the formula: $z = \frac{(X - \mu)}{\sigma}$ where X is the value of the observation, μ is the mean of the population, and σ is the standard deviation of the population.

Where to find more data to experiment on R?

- [Week 8 Exercises on the Companion](#).

I am unsure about recoding variables, I don't quite understand the steps explained in the run through. Would it be possible for there to be some sort of session or opportunity to go back through it?

- Yes, office hours.

Is it possible for you to work through more maths problems on the board, like with a calculator as if it was an exam question?

- Happy to do this in the seminar. Please ask with a concrete example, e.g. standard deviation or z-scores.

What could we be asked to do with R in the exam? Are there any example questions?

- Please wait for the [practice exam in Week 9](#).
- It opens on Monday December 2nd.

How will R actually be seen in our exams? Are we expected to be writing the code in our exam?

- You will not be using R yourself in the exam
- You are mostly interpreting R output
- Some questions ask about adjusting and debugging code

How basic do our calculators have to be for the exam?

- Please see the corresponding section [on Moodle](#) and in the [Exam FAQs](#).

Is it possible to have revision sessions during the Christmas break?

- There is no teaching outside of term
- Flo is contactable until December 19th via email for questions
- Form student-led revision groups

How can we tell when a variable is redundant? If we include it in research / an essay will it make my overall argument weaker?

- There is an argument from a conceptualisation point of view. If you do your work properly at this stage, then redundant variables should not arise.
- From a statistical point of view, redundant variables can lead to multicollinearity, which can make it difficult to interpret the results of regression analyses (PO12Q).
- Including redundant variables can also lead to overfitting, which can reduce the generalizability of the results (PO12Q).